Revisiting Property Tax Capitalization

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Abstract

This paper investigates the extent of property tax capitalization in the context of a progressive property tax pilot in Shanghai. I utilize a difference-in-differences approach by comparing neighborhoods with different tax rates before and after the implementation of the property taxes. Neighborhoods with a 0.2 percentage point higher marginal property tax rate experience a roughly 2.73% decrease in housing prices relative to their counterparts. The result reflects that at least 71% of expected property tax liabilities are capitalized into housing prices in a year. These changes also imply a large wealth redistribution as large as 2.68 years of average disposable income across homeowners.

Keywords: Property taxes; Housing prices; Capitalization
JEL: D12, H71, R21

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1 Introduction

Previous literature has shown that property taxation has a negative effect on housing prices as long as it can be at least partially capitalized (e.g. Orr (1968), Wicks et al. (1968), Smith (1970), Church (1974), Rosen and Fullerton (1977), Palmon and Smith (1998b), Kuang et al. (2012), Gallagher et al. (2013), Giertz et al. (2021)).[1] Following Tiebout (1956), most research models housing prices as the equilibrium outcome of a process in which mobile households choose among a set of communities with varying locational amenities, public service levels, and tax rates. The canonical housing bid-rent model indicates that potential homebuyers consider the value of housing services along with property taxes when choosing between alternative locations (Yinger (1982)). Theoretically, housing prices should be equal to the present discounted values of housing services net of expected tax liabilities (Oates (1969)). However, property tax is not always fully capitalized, especially when there are information asymmetries and inaccurate expectations about future tax liabilities. Understanding the extent of property tax capitalization provides insights relevant to the incidence of property taxation and rationality of agents in the real estate markets.

This paper explores the degree of property tax capitalization taking advantage of a unique property taxation in Shanghai.[2] Shanghai started to implement property taxes as a pilot to prevent housing prices from dramatically increasing on January 28, 2011. Although there might be rumors regarding the implementation of a property tax in Shanghai, the implementation details were announced without warning and the policy was implemented immediately after its announcement.[3] The prop-

[1] For a review of property tax studies, please refer to Sirmans et al. (2008).

[2] There are two cities in China, Shanghai and Chongqing, which started to implement the property taxes as pilots. Due to data availability, I focus on the policy in Shanghai.

[3] There are no implementation details revealed in the rumors before the official announcement. For example, `https://www.charltonslaw.com/newsletters/china-news-alerts/en/2010/333/Shanghai_plans_property_tax_report.pdf`. The implementation details are crucial in this context because people would not know how to correctly react to the policy without knowing whether and how they are affected. In this paper, the division of the treatment and control groups also depends on the implementation details. Even if people react to the rumors, it would not affect the identification strategy unless people in different tax groups react differently.

erty taxes in Shanghai are progressive in terms of the tax rates. Specifically, the tax rates are determined by the unit prices of the transacted properties. The tax rate for taxable houses with unit price lower than or equal to twice of the city average unit price in the last year is 0.4%, while the tax rate for those with unit price higher than twice of the city average unit price in the last year is 0.6%. This progressivity feature makes it possible to identify the degree of property tax capitalization by comparing changes in housing prices for neighborhoods subject to different tax rates.

I first introduce a simple property tax capitalization model allowing for partial capitalization. This model features two types of properties corresponding to the case in Shanghai: one with high property tax rate and the other with low property tax rate. Based on the model, the property tax capitalization rate is derived as a function of several parameters, including the two property tax rates, the discount rate, the remaining years of property ownership, and the differential percent change in housing prices for neighborhoods in the two tax brackets because of the property taxation. The last parameter is the key parameter to be estimated and other parameter values are taken from averages of economic variables.

To empirically estimate the key parameter, I obtain monthly unit housing prices for more than one thousand neighborhoods in Shanghai during 2010-2011 from Cityre Data. These neighborhoods are divided into two groups—high tax bracket and low tax bracket—based on their unit prices in 2010 because the *ex ante* unit prices could better reflect their real housing values that have not been affected by the property taxes yet. I apply a difference-in-differences (DID) approach, comparing neighborhoods in the high tax bracket and those in the low tax bracket before and after the implementation of the property taxes. I restrict the sample to neighborhoods that remain in the same tax brackets to rule out the bunching effect. I find that housing prices in the high tax bracket decrease by 2.73% relative to those in the low tax bracket due to the implementation of the property taxes. This leads to a capitalization rate of 0.71, which implies that approximately 71% of the expected property taxes could be capitalized into housing prices. In other words, current homeowners bear 71% of the tax burden while future homebuyers pay the rest 29%.[4] Further-

[4]This conclusion is based on no other private agreements between buyers and sellers on property

more, the price changes also led to a large wealth redistribution as large as 2.68 years of average disposable income per capita among homeowners in Shanghai.

To address the concern that housing markets for properties in the two tax brackets are not separable so that there might be substitution between them, I conduct robustness checks to remove neighborhoods that are potentially substitutable to one another. Specifically, I drop neighborhoods that belong to different tax groups but with similar unit prices or located close to each other. To deal with the concern that neighborhoods in the two tax brackets differ in locational amenities making them not comparable to one another, I employ a propensity score matching-difference in differences (PSM-DID) approach based on these locational amenities. Additionally, I conduct extra robustness checks to tackle potential issues, including taking announcement effect into account, using data after the housing purchase restriction, restricting the sample to neighborhoods that are unaffected by new subway openings, dropping extreme values (i.e. top and bottom 5% of housing prices), restricting the sample to a balanced panel, aggregating data into two periods, and placebo tests with falsified cutoffs and timing. The results are shown to be robust to these checks.

It is noteworthy that the degree of property tax capitalization estimated in this paper should be thought of as a lower bound for the following reasons. *First*, the special exemption rule on taxable areas complicates the effective tax rates. Since there might be an exemption for some people, the effective tax rates and their difference could be even lower than the numbers used in this paper.[5] By assuming exempted people also subject to the same tax rates, the impact of 1% change in the property tax rate is underestimated. *Second*, buyers and sellers in the resale housing markets tend to report a lower price to evade transaction taxes (Fan et al. (2022)). Similarly, they could also under-report the transacted prices to avoid the high property tax rate. Consequently, the reported prices used to determine tax liabilities could be lower than the actual transacted prices. In this paper, I use the prices close

tax payment. The homebuyers actually pay the future property taxes, but part of the burden is shared by the sellers (current homeowners) in terms of lower housing prices. The decrease in housing values is equivalent to 71% of the expected values of the future property taxes, which is implicitly paid by current homeowners.

[5]Please see Section 2 for details.

to the actual transacted prices rather than the reported prices, which over-calculates the effective tax rates hence underestimates the impact of 1% change in the property tax rate.[6] *Third*, this paper uses data at the neighborhood level while the property tax rate is applied for each housing unit. Housing units in the same neighborhoods could vary in their unit prices because of characteristics such as size, floor and orientation.[7] The mix of housing types might under-calculate the effective tax rates for expensive housing units in cheap neighborhoods and over-calculate the effective tax rates for cheap housing units in expensive neighborhoods, leading to an underestimate of the impact of 1% change in the property tax rate.[8] *Fourth*, the property taxes only target a subset of properties (i.e. local residents' newly purchased houses which are not their first property and non-citizens' all newly purchased houses) while the housing prices are averaged at the neighborhood level without distinguishing between houses subject to property taxes and those which are not. This leads to an underestimate of the property tax capitalization because the decrease in housing demand, hence the decrease in housing prices, would not be as large as the case where all potential transactions are subject to property taxes.[9]

There is no consensus on the extent of property tax capitalization in the literature. Table A1 summarizes the results of selected previous studieson property tax capitalization focusing on different locations and different time periods. The estimates vary from positive capitalization (counter results) to overcapitalization. The capitalization rate in most studies ranges from 0 to 1 (no capitalization, partial capitalization and full capitalization). My estimate is a bit larger than estimates in early studies with partial capitalization results (Oates (1969), Edel and Sclar (1974),

[6]This issue is the most severe among housing units whose unit prices are just above the cutoff. For housing units whose prices are much higher, the misreports are easier to be detected. To alleviate this problem, I remove neighborhoods whose prices are around the cutoff in one of the robustness checks.

[7]In general, housing units with smaller sizes, in the middle floors and facing the south end up with higher unit prices compared to their counterparts in the same neighborhoods.

[8]For example, the unit price for housing unit A is above the cutoff but the average unit price for the neighborhood that it belongs to is below the cutoff. In this case, housing unit A is assigned with a low property tax rate but it is actually subject to a high property tax rate. Similar to the misreports, this issue is the most severe among neighborhoods whose unit prices are around the cutoff. In one of the robustness checks, I remove neighborhoods whose prices are around the cutoff.

[9]Please see Appendix for a detailed discussion.

Stewart (1978), Krantz et al. (1982)) and is in line with that in the most recent study (Giertz et al. (2021)).

This paper contributes to the literature by providing new evidence on the degree of property tax capitalization under a unique setting with an alternative method to solve the endogeneity issues. Early studies on property tax capitalization mainly analyze local or regional cross-sectional variation in property tax rates with a hedonic pricing method. However, there are three fundamental identification problems with this approach. *First*, higher property taxes are usually associated with better public goods because higher property taxes imply higher tax revenues which can be used to finance public services (e.g. Polinsky and Rubinfeld (1978), Dye and McGuire (1997)). In the meantime, better public goods are usually associated with higher housing prices in the sorting process, making it difficult to isolate the effects of property taxes (e.g. Sieg et al. (2002), Banzhaf and Farooque (2013), Zheng et al. (2014)). *Second*, a key challenge for hedonic methods with cross-sectional variation is omitted variables. Failing to fully or accurately control for characteristics that affect housing prices leads to an omitted variable bias. *Third*, property tax rates could be endogenously determined by local governments after comprehensively considering local public good provision and housing prices. This creates a simultaneity bias between the property tax rates and housing prices when comparing across jurisdictions.

One strand of literature has used a two-stage least squares (2SLS) estimation to account for the endogenous nature of property taxes (e.g. Oates (1969), Pollakowski (1973), Church (1974), Meadows (1976), Rosen and Fullerton (1977), Chinloy (1978), Stewart (1978), Man (1995), Palmon and Smith (1998a)). However, the local average treatment effect estimated from 2SLS depends heavily on the chosen instrumental variables (or predetermined variables in early papers). Employing an alternative difference-in-differences approach, I am able to isolate the extent of property tax capitalization and that of public good capitalization by directly comparing neighborhoods with similar local public good provision but with different tax rates. Moreover, I solve the omitted variable bias by investigating both cross-sectional and temporal variation in property taxes with panel data. By adding neighborhood fixed effects and time fixed effects, I control for time-invariant neigh-

borhood characteristics and shocks common to all properties. In addition, I solve the simultaneity bias by comparing neighborhoods within a jurisdiction because the property tax rates in Shanghai are determined based on housing market conditions in the whole city but not based on any specific district or region (with more or less neighborhoods in the high or low tax groups). Combining panel data with the unique setting in Shanghai, I overcome the endogeneity issues with an alternative method and provide solid identification in estimating the capitalization rate.

Perhaps more importantly, I revisit property tax capitalization in one of the largest developing countries; most of the existing studies focus on developed countries. For example, Elinder and Persson (2017) investigates a substantial cut in the national property tax in Sweden using a similar DID approach. Among studies investigating property taxes in China, most of them focus on whether the property taxes could curb housing prices (e.g. Zheng et al. (2013), Bai et al. (2014), Du and Zhang (2015)) or use numerical simulations to analyze the potential effects of various property tax scenarios (e.g. Cao and Hu (2016), Wang et al. (2019), Zhu and Dale-Johnson (2020)). However, there is little research characterizing the degree of property tax capitalization in China. Besides, I explore the progressivity feature of the property taxes and discuss subsequent welfare implications. The findings in this paper are of significant importance as a "touchstone" as it renders us a better understanding of the property tax pilot in China, especially about the progressivity feature, so that it could help with making future policies in both modifying current property taxes and extending the property taxes into other cities or other countries with similar development stages.

The rest of the paper proceeds as follows. Section 2 introduces background and describes data. Section 3 derives the capitalization model for estimation. Section 4 presents empirical strategies and main findings. Section 5 discusses welfare implications. Section 6 concludes.

2 Background and Data

2.1 Property Taxes in Shanghai

In the past decades, China has been experiencing a dramatic increase in housing prices. The magnitude is astonishing: the national average housing price has tripled from 2005 to 2009 according to the National Bureau of Statistics of China. Consequently, houses in many Chinese cities are among the least affordable ones. According to the 11th Annual Demographia International Housing Affordability Survey in 2014, the housing affordability rating, which is the ratio of median housing price to median annual disposable household income, equals 4.2 on average for all nations. However, Economist Intelligence Unit's China Index of Housing Affordability by City reports that China has a rating of 8.6 on average, which is twice as much as the international average.[10]

Under such circumstances, as one of the pilot cities, Shanghai started to implement property taxes to curb housing prices on January 28, 2011. The policy mainly targets local residents' newly purchased houses which are not their first property and non-citizens' all newly purchased houses.[11] The tax rate for taxable houses with unit price lower than or equal to twice of the city average unit price in the last year is 0.4%, while the tax rate for those with unit price higher than twice of the city average unit price in the last year is 0.6%. The tax base is 70% of the taxable property values. Therefore, the effective tax rates for the high tax bracket and the low tax bracket are 0.42% and 0.28% respectively. Each local resident could claim a tax exemption on values associated with the first 60 square meters of all his/her houses. For example, if a homeowner has already owned a house with 50 square meters and plans to buy a new house with 100 square meters, then the taxable area for the new house is 90 square meters. The exemption applies to his/her old house

[10]Especially, Shenzhen has a rating of 19.6 (which is the least affordable city), and Beijing has a rating of 16.6, both of which are more than four times as much as the international average. Even the most affordable city, Hohhot, has a rating of 4.9, which is larger than the international average. As a comparison, the United States has a rating of 3.6, while the United Kingdom has a rating of 5.0.

[11]Local residents indicate households with Shanghai Hukou. Hukou is a system of household registration in China. A change of Hukou typically requires a job (mainly from nation-owned institutions) that is able to provide a Hukou change.

(50 square meters) and part of his/her new house (10 square meters). This exemption rule complicates the effective tax rates because it varies person by person. In the empirical section, I mainly use the difference in the two tax rates which makes it less of an issue.[12]

The property tax rate for a specific housing unit is updated on an annual basis. For each year, the tax rate for taxable houses with unit price lower than or equal to twice of the city average unit price in the last year is 0.4%, while the tax rate for those with unit price higher than twice of the city average unit price in the last year is 0.6%. Table A2 summarizes the condition for a housing unit subject to the low property tax rate (i.e. 0.4%). For example, the average unit housing price in Shanghai in 2010 is 14,213 RMB per square meter, then housing units whose transaction prices are below 28,426 (=2$\times$14,213) RMB/m^2 are subject to a 0.4% tax rate. The changing cutoffs complicate the division of the treatment and control groups. In this paper, I only focus on the impact in the first year after the implementation of the property taxes (i.e. 2011). Specifically, I divide neighborhoods into two groups based on their unit housing prices in 2010 which determine their property tax rates in 2011, assuming that the *ex ante* prices reflect their true values. The changes in housing prices in 2011 then reveal the influence of the property taxes.

A notice will be mailed to individuals who are required to pay the tax and the deadline for payment is December 31 for each year. According to the notice, in case a family fails to pay the full tax in time, an extra 0.05% late fee will be imposed on a daily basis from January 1 in the next year.[13] The property tax revenue would be used for construction of indemnificatory apartments which are public rental housing targeting low-income residents who could not afford buying their own houses. That said, local public services that are potentially beneficial to homeowners or homebuyers would not be affected by the implementation of the property taxes. Even if the government can use the money otherwise used to finance public rental apartments for public goods, it is not likely for the government to distinguish between

[12]The exemption rule generally makes the effective tax rates even lower, hence making their difference smaller. Therefore, the capitalization rate calculated in this paper can be taken as a lower bound.

[13]`http://www.china.org.cn/business/home_sweet_home/2011-02/24/content_21990132.htm`.

people who pay more and those who pay less. That is, households in a certain region, say within the same district, would all be covered, including homeowners suffering from different tax rates. This could alleviate the endogeneity issue of public good provision in prior literature.

2.2 Other Relevant Events

In this subsection, I discuss two relevant events that could potentially affect my analysis, including the housing purchase restriction and new subway openings.

Housing Purchase Restriction: Shanghai government announced a housing purchase restriction in October 2010. Each household could only purchase up to one house/apartment after the housing purchase restriction.[14] Besides, if this newly purchased house/apartment is the household's first property, then the down payment should be at least 30%; if this newly purchased house/apartment is the household's second property, then the down payment should be at least 50%; all banks stop providing loans to households for their third property. The goal of the housing purchase restriction is to prevent the irrational investment in the housing market, so it exerts extra costs for households with multiple properties. Therefore, it is possible that households prefer expensive houses/apartments to cheap ones after the housing purchase restriction.[15] If this happens, it could have impacts on relative housing prices with different values.[16] To test on this, a robustness check using data after the housing purchase restriction and a placebo test at the time of the housing purchase restriction are conducted.

New Subway Openings: There were several new subway openings during the data period, which are shown in Figure A1. For each new subway opening, I calculate a different set of distances to the nearest subway stations. New subway openings are controlled in all specifications. To further rule out the effect of new subway openings, in one of the robustness checks, I restrict the observations to only

[14]This restriction does not target houses/apartments that households have already purchased in the past.

[15]For example, I previously planned to buy 3 houses/apartments, one for living and two for investment. When the restriction came out, I might change my plan and would only buy one or two more expensive ones.

[16]This actually goes the opposite direction compared to the property taxes.

include unaffected neighborhoods which are the neighborhoods whose distances to the nearest subway stations did not change over the data period.[17]

2.3 Data

The primary dataset used in this paper is the housing prices from Cityre Data for 2010-2011.[18] Cityre Data collects list prices and sale prices from their own users (e.g. buyers and sellers), intermediaries and the Internet and improves their data quality by validating information from various sources. The dataset is constructed mainly based on list prices and adjusted by available sale prices.[19] The unit prices, measured by RMB per square meter, are averaged at the monthly neighborhood (*xiaoqu*) level.[20] A total of 1,067 neighborhoods in Shanghai are included.

The average unit housing price in Shanghai in 2010 is 14,213 RMB per square meter, which is used to calculate the cutoff for different tax rates. I divide neighborhoods into two groups based on their unit housing prices in 2010: below and equal to 28,426 RMB/m^2 which corresponds to the 0.4% marginal tax rate and above 28,426 RMB/m^2 which corresponds to the 0.6% marginal tax rate. Figure 1 shows the geographical distribution of neighborhoods available in the dataset. Neighborhoods in the low tax bracket are marked as squares in blue, while neighborhoods in the high tax bracket are marked as circles in red. Figure 2 shows the distribution of unit housing prices in 2010 and 2011 respectively. The distribution is positively skewed, with a tail of high price properties. The majority of neighborhoods are subject to a low property tax rate (i.e. 0.4%). Compared to the distribution in 2010, housing prices tend to cluster close to the cutoff. Table 1 reports the average unit housing prices before and after the implementation of property taxes for the two

[17] Although an enhanced subway system could also affect utility of subway takers and housing values, the distance to the nearest subway stations would have a more direct and vital impact. Therefore, restricting the sample to neighborhoods whose distances to the nearest subway stations did not change could largely remove the effect of new subway openings.

[18] Cityre Data cooperates with the China Real Estate Industry Association to provide information on housing markets in China. Please visit `http://www.cityre.cn`.

[19] For a discussion of the data quality, please see Lyu (2022).

[20] A neighborhood in Shanghai can accommodate up to thousands of households, so there is enough transaction information at the neighborhood level. In addition, households in the same neighborhood are highly concentrated so that amenities are similar.

tax groups in Shanghai. The summary statistics show that the lower bracket has a 6.8% increase in housing prices, while the higher bracket involves a 0.4% decrease in housing prices.

Besides, I construct a set of amenity variables for each neighborhood, including distance to the city center, distance to the nearest bus stops, number of bus lines within 500 meters and distance to the nearest subway stations.[21] The amenity data are scraped from Baidu/Google Maps API and the variables are calculated by arcGIS. During 2010-2011, there were several new subway openings in Shanghai as shown in A1. For each new subway opening, I calculate a different set of distances to the nearest subway stations. For example, Line 13 was put into use on April 20, 2010. So I calculate the neighborhoods' distances to the nearest subway stations without Line 13 for April 2010 since Line 13 was not in use for most of the days in that month, but calculate their distances to the nearest subway stations with Line 13 for May 2010. In the empirical analysis, I control for new subway openings. In one of the robustness checks, I restrict the observations to only include unaffected neighborhoods.[22] I also conduct a couple of robustness checks using these locational amenities to match neighborhoods in the high tax bracket and those in the low tax bracket and estimating the econometric model with the matched sample.

In addition, I obtain several annual district-level covariates from the China County Statistical Yearbook. The district-level population, housing investment and consumption (i.e. total retail sales of consumer goods) are included as control variables in the main regressions. The variable "total retail sales of consumer goods" includes consumption (housing not included) for non-production and non-commercial purposes and reflects an overall consumption level. I include it as a control variable because it is related to the income level which is not available at the district level.

[21] Since the bus system is more mature than subways in Shanghai, the number of bus lines within a radius also matters other than the distance to the nearest bus stops. Bus lines in different directions are counted twice because some bus stops only have buses in one direction. Subways are much less densely distributed than buses, so I only consider the distance to the nearest subway stations.

[22] A neighborhood is defined as unaffected if its distance to the nearest subway stations did not change over the data period.

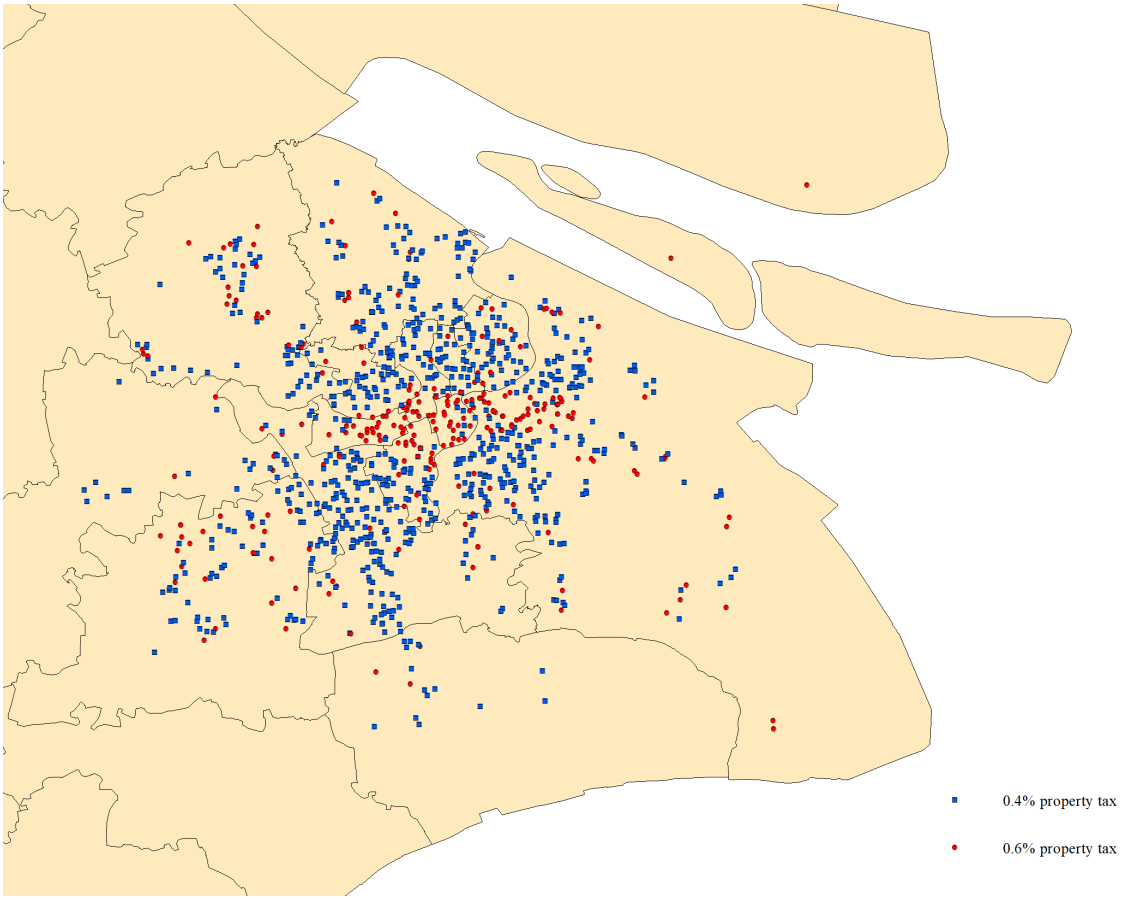

Figure 1: Neighborhoods with Different Property Tax Rates

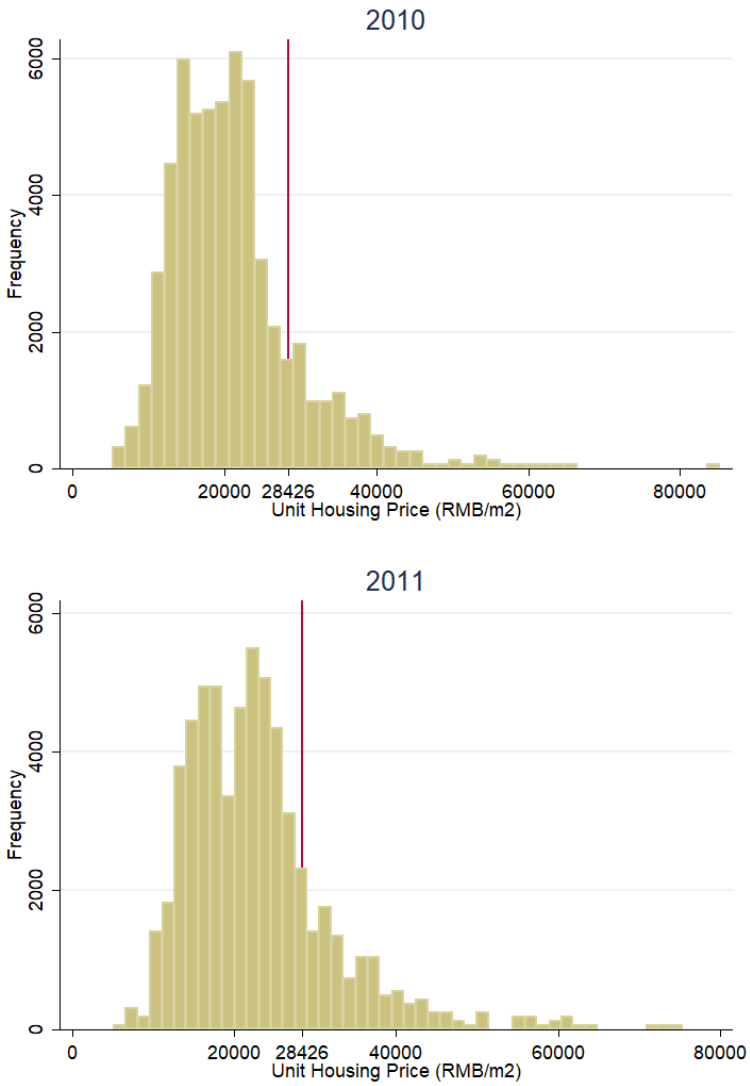

Figure 2: Distribution of Unit Housing Prices

Table 1: Average Unit Housing Prices Before and After Property Taxes by Tax Group

Marginal Tax Rate	Housing prices (RMB/m^2)				
	Percent of Observations	Pre-	Post-	Difference	Percent Change
0.4%	75%	19039.72	20340.1	1300.38	6.8%
		[4718.585]	[5159.422]	(77.657)	
0.6%	25%	36694.06	36540.21	-153.85	-0.4%
		[8319.123]	[10674.43]	(325.269)	

Note: This table reports the average unit housing prices before and after the implementation of property taxes for the two tax groups in Shanghai. Standard deviations are reported in the brackets. Standard errors are reported in the parentheses.

3 Capitalization Model

3.1 Full Capitalization

Following Tiebout (1956), most research on local public finance models housing prices as the equilibrium outcome of a process in which mobile households choose among a set of communities with varying locational amenities, public service levels, and tax rates. Property tax, as a "locational amenity" that is directly linked with housing ownership, should have strong effects on housing prices given other factors fixed. The canonical housing bid-rent model indicates that potential homebuyers consider the value of housing services along with property taxes when choosing between alternative locations (Yinger (1982)). Theoretically, housing prices should be equal to the present discounted values of housing services net of expected tax liabilities (Oates (1969)). That is, the price of a typical property is given as:

$$P = \sum_{n=0}^{N} \frac{E_n(X, Z)}{(1+r)^n} - \sum_{n=0}^{N} \frac{T_n}{(1+r)^n} \tag{1}$$

where $E_n(X, Z)$ is the pre-tax value of housing services which is a function of housing characteristics X (e.g. area, age, floor, orientation, availability of elevators, etc.) and locational characteristics Z (e.g. local public services, transportation, schools, safety, neighborhood connection, etc.), T_n is the property tax payment, r is the discount rate, N is the remaining years of property ownership.

Equation (1) suggests that the present discounted values of property tax payments should be fully capitalized into housing prices given housing services unchanged. Therefore, if two properties face different property tax rates, *ceteris paribus*, the difference in their housing prices should be exactly equal to the difference in their present discounted values of property tax payments.

3.2 Partial Capitalization

In reality, property tax is not always fully capitalized, especially when there are information asymmetries, inaccurate expectations about future tax liabilities, etc. In this case, we should allow partial capitalization, that is, there exists a capitalization parameter δ such that:

$$P = \sum_{n=0}^{N} \frac{E_n(X, Z)}{(1+r)^n} - \delta \sum_{n=0}^{N} \frac{T_n}{(1+r)^n} \tag{2}$$

If $\delta = 0$, there is no capitalization which implies that the property tax burden is fully transferred to the new homebuyers. If $0 < \delta < 1$, there is partial capitalization which indicates that the current homeowners bear δ proportion of the property tax burden while the future homebuyers take the rest of the tax burden. If $\delta = 1$, it reduces to the case of full capitalization where the current homeowners fully bear the property tax burden by receiving lower prices equivalent to the present discounted values of future tax liabilities. If $\delta > 1$, there is overcapitalization where the market overreacts so that the current homeowners bear more tax burden than the present discounted values of future tax liabilities. In rare cases, there exists $\delta < 0$, that is, positive capitalization (i.e. $-\delta > 0$), which implies that the property taxes increase housing prices instead.[23]

One complexity in measuring the degree of property tax capitalization is that the property tax revenues are usually used to finance local public goods so that housing services could hardly remain the same with changes in property taxes. The progressive property tax pilot in Shanghai provides a good context in that it is explicitly

[23]This could happen if households expect significant public service improvements (with even more values than property taxes collected).

documented that the property tax revenues in Shanghai are not used to finance local public goods. As a result, the implementation of the property taxes would not affect housing services differently for properties with different property tax rates.

Now consider a simple case where two properties initially without property taxes are exerted property taxes with different tax rates. The prices for the two properties when there are no property taxes can be written as:

$$P_0^H = \sum_{n=0}^{N} \frac{E^H}{(1+r)^n} \tag{3}$$

$$P_0^L = \sum_{n=0}^{N} \frac{E^L}{(1+r)^n} \tag{4}$$

where P_0^H is the housing price for the property that will be exerted a higher property tax, E^H is the housing service provided by this property. P_0^L is the housing price for the property that will be exerted a lower property tax, E^L is the housing service provided by this property. [24]

With the implementation of the property taxes, one property is subject to a higher property tax T^H and the other property is subject to a lower property tax T^L. Then allowing partial capitalization, their prices can be expressed as:

$$P_1^H = \sum_{n=0}^{N} \frac{E^H}{(1+r)^n} - \delta \sum_{n=0}^{N} \frac{T^H}{(1+r)^n} \tag{5}$$

$$P_1^L = \sum_{n=0}^{N} \frac{E^L}{(1+r)^n} - \delta \sum_{n=0}^{N} \frac{T^L}{(1+r)^n} \tag{6}$$

where P_1^H is the housing price for the property that are exerted a higher property tax, where P_1^L is the housing price for the property that are exerted a lower property tax. Here, we assume that the housing services provided by the two properties remain unchanged after the property taxes, which is generally a strict assumption. However, it is reasonable in the context of this study because the property tax revenues in Shanghai are not used to finance local public goods. Another important

[24] For simplicity, we assume both properties have the same remaining life.

assumption is that the higher-taxed houses share the same extent of capitalization as the lower-taxed houses, that is, I do not allow for heterogeneous capitalization in the model. Gárate and Pennington-Cross (2023) shows that the elasticity of housing supply, the price of municipal services and the type of public goods are factors that lead to heterogeneous property tax capitalization. In the context of this paper, these factors are not likely to be significantly different for higher-taxed houses and lower-taxed houses because the tax rates are not divided based on jurisdictions and the housing markets are not completely segmented.

Combing Equation (3), Equation (4), Equation (5) and Equation (6),[25] we have:

$$\begin{aligned}\delta &= -\frac{\frac{P_1^H - P_0^H}{P_0^H} - \frac{P_1^L - P_0^L}{P_0^L}}{\sum_{n=0}^{N} \frac{1}{(1+r)^n}\left(\frac{T^H}{P_0^H} - \frac{T^L}{P_0^L}\right)} \\ &= -\frac{\beta}{\theta\left(\tau^H - \tau^L\right)}\end{aligned} \tag{7}$$

where $\beta = \frac{P_1^H - P_0^H}{P_0^H} - \frac{P_1^L - P_0^L}{P_0^L} = \frac{\triangle P^H}{P_0^H} - \frac{\triangle P^L}{P_0^L}$, $\theta = \sum_{n=0}^{N} \frac{1}{(1+r)^n}$, $\tau^H = \frac{T^H}{P_0^H}$, $\tau^L = \frac{T^L}{P_0^L}$. τ^H is the higher property tax rate, τ^L is the lower property tax rate. To calculate θ, I use the average interest rate and the remaining years of property ownership during 2010-2011. The remaining challenge is to estimate β which gives the relative percent change in housing prices for two types of properties with different tax rates. In the empirical section, I apply a difference-in-differences approach to estimate the differential effect of property taxes on housing prices for the two types of properties.

[25] Taking a difference between Equation (3) and Equation (5) and divide both sides by P_0^H, we got $\frac{P_1^H - P_0^H}{P_0^H} = -\delta \sum_{n=0}^{N} \frac{\frac{T^H}{P_0^H}}{(1+r)^n}$. Taking a difference between Equation (4) and Equation (6) and divide both sides by P_0^L, we got $\frac{P_1^L - P_0^L}{P_0^L} = -\delta \sum_{n=0}^{N} \frac{\frac{T^L}{P_0^L}}{(1+r)^n}$. Then taking the difference between the two new equations, we have $\frac{P_1^H - P_0^H}{P_0^H} - \frac{P_1^L - P_0^L}{P_0^L} = -\delta \sum_{n=0}^{N} \frac{\frac{T^H}{P_0^H} - \frac{T^L}{P_0^L}}{(1+r)^n}$. This leads to Equation (7) by a simple transformation.

4 Empirical Estimation

4.1 Estimating β

To estimate β from the capitalization model, I apply a difference-in-differences approach, comparing neighborhoods in the higher tax bracket and those in the lower tax bracket before and after the implementation of the property taxes. The estimating equation is given as follows:

$$log(P_{it}) = \alpha + \beta\ 1\{P_{i,2010} > 2\bar{P}_{2010}\} \times Post_t + \gamma X_{it} + \eta_i + \theta_t + \varepsilon_{it} \quad (8)$$

where i denotes neighborhood, t denotes time (i.e. sample month), $log(P_{it})$ is the log form of unit housing prices for neighborhood i at time t, $1\{P_{i,2010} > 2\bar{P}_{2010}\}$ is an indicator which equals 1 if neighborhood i's average unit housing price in 2010 is above twice of the average unit housing price in 2010 in Shanghai, $Post_t$ is an indicator whether the property taxes have been implemented at time t, X_{it} is a vector of time-varying factors, including each neighborhood's distance to the nearest subway stations and district-level population, housing investment and consumption (total retail sales of consumer goods). η_i is the neighborhood fixed effects, θ_t is the time fixed effects, ε_{it} is the error term. β is the coefficient of interest, which captures the percent change in housing prices for neighborhoods in the higher tax bracket relative to those in the lower tax bracket due to the property taxes.

Since the property tax rate is associated with the transacted unit prices, the buyers have the incentive to negotiate with the sellers to lower the prices into the low tax bracket for neighborhoods whose prices are just above the cutoff. The sellers have the incentive to cooperate with the buyers to achieve successful transactions, and the intermediaries could also help in the negotiating process. As a result, neighborhoods whose prices are just above the cutoff would experience a decrease in order to fall into the low tax bracket, that is, it is possible that bunching at the cutoff could play an important role (Leung et al. (2014), Best and Kleven (2018)). To rule out this possibility, I restrict the sample to neighborhoods that remain in the same tax group all through this time period (2010-2011). For example, if a neighborhood's

unit housing prices are above the cutoff for all months in 2010 but its unit prices fall below the cutoff for some months in 2011, then this neighborhood should be dropped from the sample.

To identify the causal effect, there are three key assumptions required. First, the policy needs to come as a shock without any anticipation. This assumption is likely to be met because the implementation details of the property taxes were announced unexpectedly and the policy was implemented immediately after the announcement. Although there might be rumors regarding the implementation of a property tax in Shanghai, there were no implementation details revealed before its official announcement. Since the division of the treatment and control groups depends on the implementation details, it would not affect the identification strategy unless people in different tax groups react differently to the rumors. In one of the robustness checks, I estimate the announcement effect to rule out this possibility. Second, no other shocks that could have similar impacts coincided with the implementation of the policy. I have discussed several relevant events that could potentially affect my analysis in Section 2. Finally, the control group should be a good control so that the treated neighborhoods would behave similarly to the controlled neighborhoods were the policy not implemented. To test the third assumption, I conduct an event study regression as follows.

$$log(P_{it}) = \alpha + \beta_j \sum_{j=-4}^{4} 1\{P_{i,2010} > 2\bar{P}_{2010}\} \times T_{jt} + \gamma X_{it} + \eta_i + \theta_t + \varepsilon_{it} \quad (9)$$

where T_{jt} is an indicator of the time t being in the jth two-month period after the property taxes. I use the two-month period right before the property taxes as the baseline, that is, $j = -1$. A non-negative j means post-policy and a negative j means pre-policy. β_j's are the coefficients of interest. We should not expect significant estimates for negative j's which serve as a test for the parallel pre-trends. β_j's for non-negative j's give us the impacts of the property taxes on the housing prices for neighborhoods subject to high property tax rate relative to those subject to low property tax rate jth two-month period after the policy.

Table A4 presents the results of estimating Equation (9). The first two columns

report the results without control variables, and the rest two columns report the results controlling for each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption. Column (1) and (3) report the results taking two months prior to the property taxes as the baseline, and Column (2) and (4) report the results taking four months prior to the property taxes as the baseline.[26] Figure 3 plots the estimates in Column (3), suggesting small differences between neighborhoods with *ex post* low and high property tax rates before the property taxes. Their differences become significantly negative and large after the implementation of the property taxes, which indicates that neighborhoods with the *ex post* high property tax rate experience a relative drop in property values.

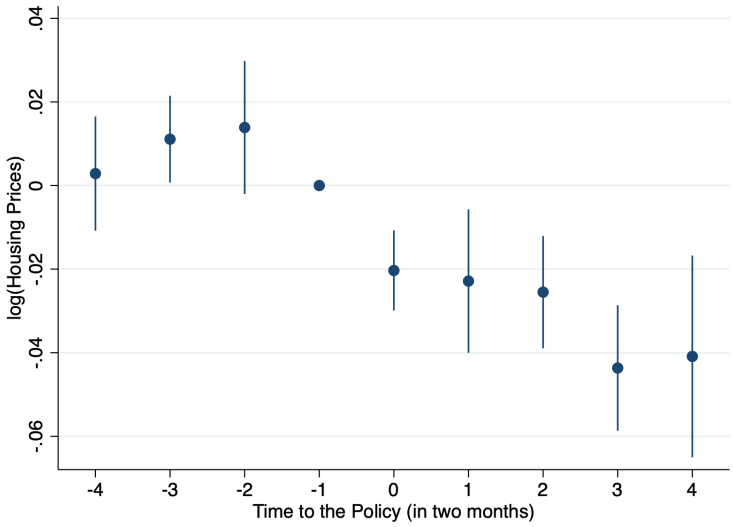

Figure 3: Event Study Estimates

Recent work has shown that using event study for pre-trends tests might subject to low power issue (e.g. Roth (2022)). Therefore, I conduct power calculations for

[26]Since two months are a relatively short period and housing prices might unexpectedly fluctuate so I extend the baseline to four months in Column (2) and (4). I still use Column (3) as my preferred specification though.

pre-trends tests in Table A3.[27] The tests suggest that if there was a linear violation of parallel trends with slope 0.00308, then we would have 50% power to detect it. By saying "detected", we mean there is a significant pre-trend coefficient. Under this slope (0.00308), the probability of passing the pre-test is 57.9% as likely as it is under parallel trends. And the realization of the pre-trends is 0.212% as likely under this slope (0.00308) as under parallel trends. Similarly, if there was a linear violation of parallel trends with slope 0.00468, then we would have 80% power to detect it. The probability of passing the pre-test under this slope (0.00468) is 23.2% and the realization of the pre-trends is 0.002% as likely as it is under parallel trends. That said, there is enough power to detect violations of parallel trends in this study.

Table 2 reports the results of estimating Equation (8). The first three columns report the results without control variables, and the last three columns report the results controlling for each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption. Column (1) and (4) report the results controlling for neighborhood fixed effects and year fixed effects, Column (2) and (5) report the results controlling for neighborhood fixed effects, year fixed effects and month fixed effects, and Column (3) and (6) report the results controlling for neighborhood fixed effects and year-month fixed effects. Standard errors are clustered at the district level to account for serial correlation in error terms among neighborhoods within a district. The results with different control variables and time fixed effects are similar across columns and the last column is my preferred specification. The estimates suggest that housing prices for neighborhoods that are assigned to the high tax bracket based on *ex ante* housing values experience a 2.73% decrease compared to those in the low tax bracket. While this estimate reflects a weighted average effect during the first year after the property taxes, the event study estimates in Figure 3 show that the largest effect after one year is approximately 4%.

[27] I use the STATA code -pretrends- for the tests. Please refer to `https://github.com/mcaceresb/stata-pretrends` for more details.

Table 2: Property Tax Capitalization

VARIABLES	log(Housing Prices)					
	(1)	(2)	(3)	(4)	(5)	(6)
High Tax Group × Post	-0.0254***	-0.0335***	-0.0267***	-0.0266***	-0.0355***	-0.0273***
	(0.00686)	(0.00722)	(0.00748)	(0.00687)	(0.00721)	(0.00770)
Dist to Subway				-0.0477***	-0.0229**	-0.00527
				(0.0111)	(0.00813)	(0.00543)
log(Population)				0.0133	0.0385	0.0174
				(0.0844)	(0.0823)	(0.0851)
log(Housing Investment)				-0.0160	-0.0105	-0.0133
				(0.0281)	(0.0273)	(0.0280)
log(Consumption)				0.0395	0.0141	0.0287
				(0.134)	(0.130)	(0.135)
Constant	10.01***	10.01***	10.01***	9.458***	9.665***	9.548***
	(0.000598)	(0.000629)	(0.000652)	(1.588)	(1.539)	(1.597)
Observations	10,100	10,100	10,100	10,100	10,100	10,100
R-squared	0.975	0.979	0.984	0.975	0.979	0.984
Neighborhood FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	NO	YES	YES	NO
Month FEs	NO	YES	NO	NO	YES	NO
Year-Month FEs	NO	NO	YES	NO	NO	YES

Note: Column (1)-(3) report the results without control variables, Column (4)-(6) report the results controlling for each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption. Column (1) and (4) report the results controlling for neighborhood fixed effects and year fixed effects, Column (2) and (5) report the results controlling for neighborhood fixed effects, year fixed effects and month fixed effects, Column (3) and (6) report the results controlling for neighborhood fixed effects and year-month fixed effects. Standard errors clustered by district are reported in the parentheses.
*** p<0.01, ** p<0.05, * p<0.1

Table 3 reports the results with longer post-treatment periods. The first column reports the result using data for 2010-2012, the next column reports the result using data for 2010-2013, and the last column reports the result using data for 2010-2014. It is noteworthy that the cutoff for different tax groups changes year by year based on twice of the city average unit price in the previous year. The city average unit prices for 2011, 2012 and 2013 are 13,448 RMB/m^2, 13,870 RMB/m^2 and 16,192 RMB/m^2 respectively. Therefore, the cutoffs for 2012, 2013 and 2014 are 26,896

RMB/m^2, 27,740 RMB/m^2 and 32,384 RMB/m^2 respectively. I keep neighborhoods whose annual average unit prices remain in the same tax brackets every year, that is, neighborhoods that switch between the two tax groups are removed. The results are consistent with the main specification, indicating a relatively stable impact of property taxes.

Table 3: Property Tax Capitalization: Longer Periods

	log(Housing Prices)		
VARIABLES	2010-2012	2010-2013	2010-2014
	(1)	(2)	(3)
High Tax Group × Post	-0.0316**	-0.0362***	-0.0337**
	(0.0126)	(0.00715)	(0.0118)
Observations	26,405	35,339	37,666
R-squared	0.970	0.963	0.962
Neighborhood FEs	YES	YES	YES
Year-Month FEs	YES	YES	YES

Note: Column (1) reports the result using data for 2010-2012, Column (2) reports the result using data for 2010-2013, Column (3) reports the result using data for 2010-2014. Neighborhood fixed effects, year-month fixed effects, each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption are controlled for all columns. Standard errors clustered by district are reported in the parentheses.
*** p<0.01, ** p<0.05, * p<0.1

4.2 Robustness

One concern underlying the identification strategy is that the housing markets for properties with different tax rates are not completely segmented. The higher tax rate on the more expensive properties might induce some prospective homebuyers to switch to the cheaper properties instead. Therefore, the cheaper segment might have experienced higher housing prices as a result of the progressive property taxes. To alleviate this issue, I use subsamples with neighborhoods that are more segmented.

When homebuyers choose between alternative options, properties with similar unit prices or those located close to each other are more likely to be substitutes. On the contrary, people are not very likely to purchase a property with completely different locations or being much more expensive or cheaper than what they desire simply because of the property taxes. Therefore, I conduct two batteries of robustness checks dropping neighborhoods that belong to different tax groups but with similar unit prices and located close to each other respectively.

Table 4 reports the results using subsamples dropping neighborhoods that belong to different tax groups but with similar unit prices. In Column (1), I drop neighborhoods whose unit prices in 2010 are 1,000 RMB/m^2 more or less around the cutoff (i.e. 28,426 RMB/m^2). That is, only neighborhoods with *ex ante* unit prices below 27,426 RMB/m^2 or above 29,426 RMB/m^2 are considered. Similarly, Column (2)-(4) show results dropping neighborhoods whose *ex ante* unit prices are 2,000 RMB/m^2, 5,000 RMB/m^2 and 10,000 RMB/m^2 more or less around the cutoff respectively. The estimates suggest that the effect becomes slightly larger but still similar to the main specification.

Table 5 reports the results using subsamples dropping neighborhoods that belong to different tax groups but located close to each other. From Figure 1, we can see that some neighborhoods with high property tax rates are clustered while others are located close to neighborhoods with low property tax rates. Using arcGIS, I create buffers of 200m, 500m, 1km and 1.5km around each neighborhood and generate dummies indicating whether there are neighborhoods from the other tax group in the buffers. In Column (1), I drop a neighborhood if there is at least one neighborhood from the other tax group in its 200m buffer. That is, only neighborhoods located more than 200m away from all neighborhoods in the other tax group are considered. Similarly, Column (2)-(4) show results dropping neighborhoods located within 500m, 1km and 1.5km of neighborhoods in the other tax group respectively. The estimates suggest that the effect becomes slightly larger but still similar to the main specification. Therefore, substitution between the two tax groups are not the deciding factor that causes the differential effect of the property taxes.

Another critical concern is that neighborhoods in the two tax brackets vary in locations, hence locational amenities, as shown in Figure 1, which might lead to

differential effect of the property taxes. Panel A in Table A5 shows that neighborhoods differ in distance to the city center, distance to the nearest bus stops and number of bus lines within 500 meters. To make sure that neighborhoods in the two groups are more comparable, I first employ propensity score matching (PSM) and then estimate Equation (8) with the matched sample. I employ the nearest neighbor matching based on neighborhood amenities (i.e. distance to the city center, distance to the nearest bus stops, number of bus lines within 500 meters and distance to the nearest subway stations) with the pre-treatment data on the common support. Panel B and Panel C in Table A5 shows that neighborhoods after matching become more similar to each other in terms of these locational amenities.[28]

Table 6 reports the results applying PSM-DID. Column (1) reports the result using sample after PSM based on a Probit model without replacement, Column (2) reports the result using sample after PSM based on a Probit model with replacement, Column (3) reports the result using sample after PSM based on a Logit model without replacement, Column (4) reports the result using sample after PSM based on a Logit model with replacement. The estimates show similar effect to the main specification, implying that neighborhoods with different locational amenities have similar trends in housing prices so that DID could remove the effect of these differences.

To address other potential concerns, I conduct a set of extra robustness checks in Table A6. First, there might be an announcement effect even if the property taxation was announced at the very end of January 2011. To rule out this possibility, I define January 2011 as the announcement period and re-estimate the regressions including the announcement period. Column (1) shows that the announcement effect is small and insignificant, especially compared with the magnitude of the main effect.

Second, I restrict the sample to time periods after the housing purchase restriction to rule out the impact of the purchase restriction. Column (2) shows that the estimate becomes slightly larger but still similar to the main specification, indicating that the purchase restriction is not the main reason to explain the effect.

Third, there are several new subway openings during 2010-2011 as shown in Figure A1, which might affect housing prices for neighborhoods around these new

[28]Locational amenities after matching with replacement are available upon request.

subways differently. To address this concern, I restrict the sample to neighborhoods that are unaffected by new subway openings during this time period (2010-2011). A neighborhood is defined as unaffected if its distance to the nearest subway stations does not change over time. Column (3) shows that the estimate is consistent with the main specification, suggesting that controlling for distance to subways in the main specification is fairly enough in this context.

Fourth, one may worry that the results are driven by neighborhoods with extremely high or low unit prices. Column (4) reports the result dropping sample with the top and bottom 5% of housing prices. The estimate is consistent with the main specification, ruling out the possibility that the effect is driven by extreme values.

Fifth, I restrict the sample to a balanced panel to deal with the concern that the missing values are endogenously dropped. Column (5) shows that the estimate becomes slightly smaller but still similar to the main specification. This implies that the effect is not driven by endogenous missing values.

Sixth, recent studies show that two way fixed effects in a DID design with multiple groups and time periods might subject to negative weights on some observations (e.g. De Chaisemartin and d'Haultfoeuille (2020), Imai and Kim (2021)). To address this concern, I aggregate the observations into two periods (one period before and one period after the property taxes) and re-estimate the regressions with only neighborhood fixed effects. Without the year-month fixed effects, the policy effect for the baseline can therefore be estimated. Column (6) in Table A6 shows that the main coefficient is consistent with the main specification. The coefficient for the policy dummy indicates that the property taxes (or other shocks that affect the whole economy coinciding with the property taxes) lead to a 8.21% increase in housing prices for neighborhoods in the low tax group. However, it is difficult to disentangle the policy effect from the time trend so that I do not claim causality for this coefficient.

Finally, I conduct several placebo tests and report the results in Table A7. I first consider two "placebo" cutoffs: one is the median price in the low tax group and the other is the median price in the high tax group. In Column (1), I restrict the sample to neighborhoods that are always in the low tax bracket and use the median price in this group as the falsified cutoff. In Column (2), I restrict the sample to neighbor-

hoods that are always in the high tax bracket and use the median price in this group as the falsified cutoff. The estimates for both placebo tests are small and insignificant, suggesting that the effect only exists at the right cutoff of the property taxes. I then consider a "placebo" timing: the timing of the housing purchase restriction. In Column (3), I use the timing of the housing purchase restriction (i.e. October 2010) as the falsified timing and consider a symmetric time window around it. The estimate is also small and insignificant, suggesting that the effect only exists at the right timing of the property taxes. Above all, the estimate for β is robust across various specifications and placebo tests.

Table 4: Property Tax Capitalization: Drop Substitutable Neighborhoods with Similar Prices

VARIABLES	log(Housing Prices)			
	Drop Neighborhoods Price $\pm$1,000 RMB/m^2	Drop Neighborhoods Price $\pm$2,000 RMB/m^2	Drop Neighborhoods Price $\pm$5,000 RMB/m^2	Drop Neighborhoods Price $\pm$10,000 RMB/m^2
	(1)	(2)	(3)	(4)
High Tax Group $\times$ Post	-0.0273***	-0.0283***	-0.0335***	-0.0414***
	(0.00770)	(0.00780)	(0.00876)	(0.00964)
Observations	10,100	10,061	9,180	4,343
R-squared	0.984	0.984	0.984	0.988
Neighborhood FEs	YES	YES	YES	YES
Year-Month FEs	YES	YES	YES	YES

Note: Column (1) reports the result using a subsample during 2010-2011 removing neighborhoods with *ex ante* unit housing prices within 1,000 RMB per square meter around the cutoff, Column (2) reports the result using a subsample during 2010-2011 removing neighborhoods with *ex ante* unit housing prices within 2,000 RMB per square meter around the cutoff, Column (3) reports the result using a subsample during 2010-2011 removing neighborhoods with *ex ante* unit housing prices within 5,000 RMB per square meter around the cutoff, Column (4) reports the result using a subsample during 2010-2011 removing neighborhoods with *ex ante* unit housing prices within 10,000 RMB per square meter around the cutoff. Neighborhood fixed effects, year-month fixed effects, each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption are controlled for all columns. Standard errors clustered by district are reported in the parentheses.
*** p<0.01, ** p<0.05, * p<0.1

Table 5: Property Tax Capitalization: Drop Substitutable Neighborhoods with Adjacent Locations

VARIABLES	log(Housing Prices)			
	Drop Neighborhoods Other Group within 200m	Drop Neighborhoods Other Group within 500m	Drop Neighborhoods Other Group within 1km	Drop Neighborhoods Other Group within 1.5km
	(1)	(2)	(3)	(4)
High Tax Group × Post	-0.0259***	-0.0248**	-0.0325**	-0.0335*
	(0.00839)	(0.00859)	(0.0139)	(0.0179)
Observations	9,684	9,089	7,372	5,226
R-squared	0.983	0.983	0.981	0.977
Neighborhood FEs	YES	YES	YES	YES
Year-Month FEs	YES	YES	YES	YES

Note: Column (1) reports the result using a subsample during 2010-2011 removing neighborhoods located within 200m of neighborhoods in the other tax group, Column (2) reports the result using a subsample during 2010-2011 removing neighborhoods located within 500m of neighborhoods in the other tax group, Column (3) reports the result using a subsample during 2010-2011 removing neighborhoods located within 1km of neighborhoods in the other tax group, Column (4) reports the result using a subsample during 2010-2011 removing neighborhoods located within 1.5km of neighborhoods in the other tax group. Neighborhood fixed effects, year-month fixed effects, each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption are controlled for all columns. Standard errors clustered by district are reported in the parentheses.

*** p<0.01, ** p<0.05, * p<0.1

Table 6: Property Tax Capitalization: PSM-DID

VARIABLES	log(Housing Prices)			
	PSM w/o Replacement Probit Model	PSM w/ Replacement Probit Model	PSM w/o Replacement Logit Model	PSM w/ Replacement Logit Model
	(1)	(2)	(3)	(4)
High Tax Group × Post	-0.0328***	-0.0319***	-0.0327***	-0.0307***
	(0.00729)	(0.00828)	(0.00672)	(0.00781)
Observations	4,148	4,124	4,148	4,148
R-squared	0.987	0.988	0.987	0.986
Neighborhood FEs	YES	YES	YES	YES
Year-Month FEs	YES	YES	YES	YES

Note: The propensity scores are calculated based on neighborhood locational amenities (i.e. distance to the city center, distance to the nearest bus stops, number of bus lines within 500 meters and distance to the nearest subway stations). Column (1) reports the result using sample after PSM based on a Probit model without replacement, Column (2) reports the result using sample after PSM based on a Probit model with replacement, Column (3) reports the result using sample after PSM based on a Logit model without replacement, Column (4) reports the result using sample after PSM based on a Logit model with replacement. Neighborhood fixed effects, year-month fixed effects, each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption are controlled for all columns. Standard errors clustered by district are reported in the parentheses.
*** p<0.01, ** p<0.05, * p<0.1

4.3 Calculating δ

Since β has been estimated using a DID approach, the next step is to calculate the capitalization rate δ. I use $\beta = -2.73\%$ from Column (6) in Table 2 which is my preferred specification. The effective property tax rates above and below the cutoff are 0.42% (=$0.6\% \times 70\%$) and 0.28% (=$0.4\% \times 70\%$) respectively. Taking the exemption rule on taxable areas into account, the effective tax rates are even lower, making their difference smaller too. The capitalization rate calculated in this paper can be hence considered as a lower bound. For the discount rate r, I use the average annual deposit interest rate during 2010-2011 which is approximately 3%.[29] The remaining years of property ownership N is calculated by total years of property ownership minus current housing age. The residential property ownership in China lasts for 70 years.[30] According to the China Population Census in 2020,

[29] https://www.reuters.com/article/idCNL3S12N4F420151023.

[30] Actually, homeowners only have the right to reside in the properties during the 70 years while the land always belongs to the nation.

5.8% of residential buildings were built before 1980s, 16.5% of residential buildings were built in 1980s, 28.0% of residential buildings were built in 1990s, 28.8% of residential buildings were built in 2000s, and 20.8% of residential buildings were built in 2010s in Shanghai.[31] I remove residential buildings built in 2010s and reweigh the rest, getting an average housing age as of 2010 roughly 15 years old. Thus, the average remaining years of property ownership as of 2010 in Shanghai is 55 years. Based on the formula $\theta = \sum_{n=0}^{N} \frac{1}{(1+r)^n}$, θ is calculated to be 27.6. Therefore, δ is calculated to be 0.71 in accord with Equation (7). Scenario 1 in Table 7 summarizes these parameter values, suggesting that 71% of property taxes could be capitalized into housing prices. That said, current homeowners bear 71% of the tax burden while future homebuyers pay the rest 29%. Alternatively, I use the largest δ after one year (i.e. -4%) in Scenario 2 and the result implies full capitalization. In Scenario 3 and Scenario 4, I assume that the property ownership lasts forever. The results show that the capitalization rate is at least 57%.

Table 7: Capitalization Parameter

Parameter	Description/Formula	Scenario 1	Scenario 2	Scenario 3	Scenario 4
β	Estimate from Equation (8)	-2.73%	-4%	-2.73%	-4%
τ^H	Effective property tax rate above cutoff	0.42%	0.42%	0.42%	0.42%
τ^L	Effective property tax rate below cutoff	0.28%	0.28%	0.28%	0.28%
r	Average annual deposit interest rate	3%	3%	3%	3%
N	Average years of remaining housing life	55	55	∞	∞
θ	$\sum_{n=0}^{N} \frac{1}{(1+r)^n}$	27.6	27.6	34.5	34.5
δ	$-\frac{\beta}{\theta(\tau^H - \tau^L)}$	0.71	1.04	0.57	0.83

5 Welfare Implications

The progressive property taxes in Shanghai have shown to result in a larger decrease in housing prices for neighborhoods with *ex ante* higher values. This finding indicates that homeowners who bought houses with higher prices before the prop-

[31] `https://new.qq.com/rain/a/20220628A0BO3R00`.

erty taxes experience a loss in their property assets relative to their counterparts. $\delta = 0.71$ implies that current homeowners bear most of the tax burden. Since residential properties are the most valuable assets for many Chinese families, changes in housing values may largely change the total wealth of these families.[32] Therefore, the progressive property taxes can lead to an implicit wealth redistribution among homeowners. I calculate the housing value changes of the high tax bracket relative to the low tax bracket by the following formula:

$$\triangle V = \triangle \% P * \bar{P} \tag{10}$$

where $\triangle \% P$ is the relative percent change in unit housing price estimated in the empirical section for neighborhoods in the high tax bracket, $\bar{P}$ is the average housing value for a typical house/apartment of 100 square meters for neighborhoods in the high tax bracket before the implementation of the property taxes, $\triangle V$ is the relative housing value change for neighborhoods in the high tax bracket due to property taxes which is equal to the product of $\triangle \% P$ and $\bar{P}$.

According to Table 2, housing prices in the high tax bracket decrease by 2.73%, that is, $\triangle \% P = -2.73\%$. Since the average unit housing price in the high tax bracket before the property taxes is approximately 35.3 thousand RMB in the dataset, the average housing value is roughly 3.53 million RMB for a typical house/apartment of 100 square meters in the high tax bracket, that is, $\bar{P} = 3.53M$. By a simple calculation with Equation (10), homeowners who bought houses with *ex ante* higher values lose 96.37 thousand RMB relative to those who bought houses with *ex ante* lower values for a typical house/apartment of 100 square meters. The magnitude of the redistribution is approximately 2.68 years of the average disposable income per capita in Shanghai in 2011 as shown in Table 8.[33] The redistribution from homeowners who bought houses with *ex ante* higher values to homeowners who bought houses with *ex ante* lower values are shown to be tremendous compared to the average disposable income per capita. This is consistent with the progressivity feature

[32] According to the China Household Wealth Survey Report 2018, housing property accounts for 69.7% of the family wealth in urban areas. See `http://www.gov.cn/xinwen/2018-12/28/content_5352858.htm`.

[33] The average disposable income per capita in Shanghai in 2011 is approximately 36 thousand RMB.

in that the property taxes exert more tax burden on homeowners who have bought houses with higher values.

Table 8: A Back-of-the-Envelope Calculation of Implicit Wealth Redistribution

Neighborhoods	$\triangle\%P$	$\bar{P}(Million\ RMB)$	$\triangle V(Thousand\ RMB)$	$\frac{\triangle V}{I}(years)$
High Tax Bracket	-2.73%	3.53	-96.37	-2.68
Low Tax Bracket	Baseline			

Note: The average disposable income per capita in Shanghai in 2011 is approximately 36,000 RMB. $I = 36$ thousand RMB/year is used to calculate $\frac{\triangle V}{I}$.

6 Conclusion

This paper investigates the extent of property tax capitalization taking advantage of a unique property tax pilot in Shanghai. Utilizing the progressivity feature of the property taxation, I estimate the capitalization rate by comparing changes in housing prices for neighborhoods subject to different tax rates. Using monthly unit housing prices for more than one thousand neighborhoods in Shanghai during 2010-2011 from Cityre Data and applying a difference-in-differences approach, I find that housing prices in the high tax bracket decrease by 2.73% relative to those in the low tax bracket due to the implementation of the property taxes. This leads to a lower bound of the capitalization rate as 0.71, which implies that at least 71% of the expected property taxes could be capitalized into housing prices. The results are consistent across several robustness checks and placebo tests. The price changes also led to a large wealth redistribution as large as 2.68 years of average disposable income per capita among homeowners in Shanghai.

This paper complements the literature by providing new evidence on the degree of property tax capitalization under a unique setting which could potentially overcome the endogeneity issues in the previous research. Moreover, I revisit property tax capitalization in one of the largest developing countries; most of the existing studies focus on developed countries. Finally, I explore the progressivity feature of the property taxes and discuss subsequent welfare implications. The property taxes implicitly transfer wealth from people owning houses with relatively higher

values to those owning houses with relatively lower values. Given that the rich are more likely to buy more expensive houses, the wealth redistribution tend to decrease inequality. This is consistent with the design of a progressive tax regime. The findings in this paper are relevant to policy as it renders us a better understanding of the property taxation incidence in China so that it could help with making future policies in both modifying current property taxes and extending the property taxes into other cities or other countries with similar development stages.

There are still some caveats that merit future work. First, the special exemption rule on taxable areas and market participants' tax evasion behaviors complicate the effective tax rates so that the capitalization rate calculated in this paper can be taken as a lower bound. The estimation will be more accurate if deductible in housing areas could be taken into consideration.[34] Second, the dataset used in this paper is at the neighborhood level which is lack of dwelling characteristics such as dwelling area, floor, orientation and number of rooms. Using housing unit transaction data would better capture the effect of the property taxes. Third, this paper does not take difficulty in property tax collection into consideration. As reported by the Shanghai Municipal Bureau of Local Taxation, the property tax revenue in Shanghai in 2011 is 7.37 billion RMB, which only comprises 2.32% of total tax revenue in 2011.[35] The low property tax revenue is inconsistent with the taxable houses. The capitalization rate is expected to be even larger taking this into account. Finally, the pilot program in Shanghai is very different from those in most other economies or even China's future property tax system in that the funding is not used directly in the public service or amenity of the neighborhoods. If the quality of life in the neighborhood is, to some extent, linked with the property taxes and homeowners can enjoy additional benefits from the property taxes, the property tax capitalization would be different from the results in the current paper.[36] Future studies that focus on the above issues will be helpful in understanding the property taxes in developing

[34] I am not able to deal with it due to data limitations.

[35] This situation gets better. During 2017-2020, this number increased to 20 billion (3.4% of total tax revenue).

[36] If public service is associated with property taxes, households would have expectations on the levels of public goods which could be higher, lower or exactly the same as the real numbers. Different expectations will affect the degree of property tax capitalization.

countries.

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A Appendix

A.1 Discussion on Treatment and Data Issues

One critical concern regarding the treatment of the property taxes in Shanghai is that the property taxes only target a subset of properties (i.e. local residents' newly purchased houses which are not their first property and non-citizens' all newly purchased houses). I demonstrate below that the affected properties and homebuyers constitute a large proportion of potential transactions and population.

According to the Shanghai Municipal Tax Service, there were 50,000 newly transacted housing units that should be subject to the property taxes as of November 2012, which was 20% of newly purchased houses in Shanghai. However, it is noteworthy that these are transactions that still happened after the property taxes. Some other potential transactions just failed due to the property taxes. The potential transactions that were targeted by the property taxes should be larger. According to the China Index Academy, the number of housing transactions dropped by 46% after three months of the property tax implementation in Shanghai. If we assume that these are potential transactions failed by the property taxes, the potential transactions targeted by the property taxes should be 56.8% (=46% +54%$\times$20%) of total potential transactions.

From the perspective of affected population, there are two groups of potential buyers who could be affected by property taxes: non-citizens who plan to buy properties and citizens who plan to buy multiple properties. First, non-citizens consist of 39% of long-term residents in Shanghai according to the national census in 2010. Second, the homeownership rate among citizens in Shanghai is 67.9% according to the China Household Finance Survey in 2012 (Earlier data in 2010 are unfortunately unavailable). That said, the potentially affected population accounts for 80.4% (39%+61%$\times$67.9%) of long-term residents in Shanghai.

An associated issue is that the housing prices reflect all purchased houses. This mixture would lead to an underestimate of the property tax capitalization. An increase in user cost due to the property taxes leads to a decrease in housing demand (for any given price) which in turn results in lower housing prices. If only partial potential homebuyers are subject to a user cost increase, the housing demand

decrease would not be that large hence the housing price drop would be smaller. This leads to a smaller capitalization rate. However, it is difficult to accurately estimate to what extent this number is underestimated because it involves estimating the slope of housing supply, which is beyond the scope of this paper. Therefore, I claim that the estimate in this paper can be taken as a lower bound.

A.2 Additional Figures and Tables

Figure A1: New Subway Openings in Shanghai (2010-2011)

Table A1: Selected Studies on Property Tax Capitalization

Author	Location	Time Period	Method	Results
Oates (1969)	New Jersey	1960	2SLS	Partial Capitalization (66%)
Pollakowski (1973)	San Francisco (CA)	1960	2SLS	No Significant Capitalization
Church (1974)	Martinez (CA)	1967-1970	2SLS	Overcapitalization
Edel and Sclar (1974)	Boston (MA)	1930-1970	OLS	Partial Capitalization (50%)
Wales and Wiens (1974)	Surrey (England)	1972	OLS	No Significant Capitalization
Stewart (1978)	Ann Arbor (MI)	1970	2SLS	Partial Capitalization (<60%)
Johnson and Lea (1982)	Erie (NY)	1978	2SLS	No Significant Capitalization
Krantz et al. (1982)	Six Pennsylvania Cities	1979	MLE	Partial Capitalization (60%)
Man (1995)	Phoenix (AZ)	1985-1988	2SLS	Full Capitalization
Palmon and Smith (1998a)	Houston (TX)	1989	2SLS	Full Capitalization
Brasington (2001)	Six Ohio Metro Areas	1991	OLS&IV	Positive Capitalization
McDonald and Yurova (2006)	Chicago (IL)	2001-2004	OLS	Full Capitalization
Gallagher et al. (2013)	Cook (IL)	2000	OLS	Full Capitalization
Elinder and Persson (2017)	Sweden	2006-2008	DID	No Significant Capitalization
Giertz et al. (2021)	Dallas (TX)	2014-2016	OLS	Partial Capitalization (>70%)

Table A2: Condition for Low Property Tax Rate by Year

Year	Average unit housing price last year (RMB/m^2)	Condition for 0.4% tax rate
2011	14213	Unit housing price $\leq$ 28426 RMB/m^2
2012	13448	Unit housing price $\leq$ 26896 RMB/m^2
2013	13870	Unit housing price $\leq$ 27740 RMB/m^2
2014	16192	Unit housing price $\leq$ 32384 RMB/m^2
2015	16415	Unit housing price $\leq$ 32830 RMB/m^2
2016	21501	Unit housing price $\leq$ 43002 RMB/m^2
2017	25910	Unit housing price $\leq$ 51820 RMB/m^2
2018	24866	Unit housing price $\leq$ 49732 RMB/m^2
2019	28981	Unit housing price $\leq$ 57962 RMB/m^2
2020	32926	Unit housing price $\leq$ 65852 RMB/m^2
2021	36741	Unit housing price $\leq$ 73482 RMB/m^2
2022	40974	Unit housing price $\leq$ 81948 RMB/m^2

Table A3: Power Test for Pre-trends

Parameter	Description	Power = 0.5	Power = 0.8
Hypothesized Trend	The slope of a linear violation of parallel trends that a pre-trends test would detect a specified fraction of the time.	0.00308	0.00468
Bayes Factor	The ratio of the probability of "passing" the pre-test under the hypothesized trend relative to under parallel trends.	0.579	0.232
Likelihood Ratio	The ratio of the likelihood of the observed pre-treatment coefficients under the hypothesized trend relative to under parallel trends.	0.00212	0.00002

Table A4: Event Study Estimates

VARIABLES	log(Housing Prices)			
	(1)	(2)	(3)	(4)
7-8 Months Prior	0.00298	-0.00395	0.00285	-0.00401
	(0.00633)	(0.00595)	(0.00633)	(0.00597)
5-6 Months Prior	0.0112**	0.00424	0.0111**	0.00422
	(0.00471)	(0.00688)	(0.00480)	(0.00694)
3-4 Months Prior	0.0140*		0.0139*	
	(0.00742)		(0.00737)	
1-2 Months After	-0.0233***	-0.0302***	-0.0203***	-0.0271***
	(0.00384)	(0.00421)	(0.00444)	(0.00562)
3-4 Months After	-0.0258***	-0.0327***	-0.0229**	-0.0297***
	(0.00658)	(0.00780)	(0.00795)	(0.00947)
5-6 Months After	-0.0285***	-0.0354***	-0.0255***	-0.0323***
	(0.00671)	(0.00610)	(0.00621)	(0.00632)
7-8 Months After	-0.0465***	-0.0534***	-0.0437***	-0.0505***
	(0.00713)	(0.00651)	(0.00695)	(0.00701)
9-10 Months After	-0.0436***	-0.0506***	-0.0409***	-0.0477***
	(0.0102)	(0.00980)	(0.0112)	(0.0113)
subway			0.0761	0.0763
			(0.0675)	(0.0676)
log(Population)			-0.0460	-0.0463
			(0.0774)	(0.0774)
log(Housing Investment)			-0.0133	-0.0133
			(0.0221)	(0.0221)
log(Consumption)			0.0557	0.0556
			(0.116)	(0.115)
Constant	10.03***	10.03***	9.410***	9.414***
	(0.000619)	(0.000593)	(1.377)	(1.377)
Observations	7,586	7,586	7,586	7,586
R-squared	0.987	0.987	0.987	0.987
Neighborhood FEs	YES	YES	YES	YES
Year-Month FEs	YES	YES	YES	YES

Note: Column (1) and (2) report the results without control variables, Column (3) and (4) report the results controlling for each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption. Column (1) and (3) report the results taking two months prior to the property taxes as the baseline, Column (2) and (4) report the results taking four months prior to the property taxes as the baseline. Neighborhood fixed effects and year-month fixed effects are controlled for all columns. Standard errors clustered by district are reported in the parentheses.

*** p<0.01, ** p<0.05, * p<0.1

Table A5: Locational Amenities for Unmatched and Matched Samples

Variable	(1) Low Tax Rate N	Mean/(SE)	(2) High Tax Rate N	Mean/(SE)	(1)-(2) Pairwise t-test N	Mean difference
A. Before Matching						
Distance to City Center (km)	804	9.167	263	8.065	1067	1.103***
		(0.168)		(0.371)		(0.362)
Distance to Nearest Bus Stops (km)	804	0.270	263	0.339	1067	-0.069**
		(0.014)		(0.033)		(0.031)
Bus Lines within 500m	804	17.231	263	20.989	1067	-3.757**
		(0.692)		(1.458)		(1.469)
Distance to Nearest Subway Stations (km)	804	1.480	263	1.617	1067	-0.137
		(0.060)		(0.165)		(0.142)
B. After Matching (w/o Replacement, Probit Model)						
Distance to City Center (km)	246	8.259	246	8.390	492	-0.131
		(0.340)		(0.383)		(0.513)
Distance to Nearest Bus Stops (km)	246	0.350	246	0.356	492	-0.007
		(0.036)		(0.035)		(0.050)
Bus Lines within 500m	246	20.825	246	19.972	492	0.854
		(1.455)		(1.514)		(2.100)
Distance to Nearest Subway Stations (km)	246	1.584	246	1.689	492	-0.105
		(0.141)		(0.175)		(0.223)
C. After Matching (w/o Replacement, Logit Model)						
Distance to City Center (km)	246	8.278	246	8.390	492	-0.112
		(0.336)		(0.383)		(0.509)
Distance to Nearest Bus Stops (km)	246	0.341	246	0.356	492	-0.016
		(0.036)		(0.035)		(0.050)
Bus Lines within 500m	246	20.589	246	19.972	492	0.618
		(1.460)		(1.514)		(2.104)
Distance to Nearest Subway Stations (km)	246	1.648	246	1.689	492	-0.041
		(0.143)		(0.175)		(0.226)

*** p<0.01, ** p<0.05, * p<0.1

Table A6: Property Tax Capitalization: Robustness Checks

VARIABLES	log(Housing Prices)					
	Announcement Effect	Post-Purchase Restriction	Neighborhoods w/o New Subways	Drop 5% Extreme Values	Balanced Panel	Aggregate Two Periods
	(1)	(2)	(3)	(4)	(5)	(6)
High Tax Group × Post	-0.0282***	-0.0330***	-0.0275***	-0.0270***	-0.0246***	-0.0323***
	(0.00812)	(0.00574)	(0.00817)	(0.00716)	(0.00757)	(0.0105)
High Tax Group × Announce	-0.00810					
	(0.00504)					
Post						0.0821***
						(0.0209)
Observations	10,100	6,331	9,055	9,090	9,744	856
R-squared	0.984	0.987	0.983	0.977	0.983	0.995
Neighborhood FEs	YES	YES	YES	YES	YES	YES
Year-Month FEs	YES	YES	YES	YES	YES	NO

Note: Column (1) reports the announcement effect (i.e. January 2011), Column (2) reports the result using data after the housing purchase restriction, Column (3) reports the result for neighborhoods that are not affected by new subway openings, Column (4) reports the result dropping sample with top and bottom 5% of housing prices, Column (5) reports the result using a balanced panel, Column (6) reports the result aggregating data into two periods: before and after the implementation of the property taxes. Neighborhood fixed effects, year-month fixed effects (except Column (6)), each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption are controlled for all columns. Standard errors clustered by district are reported in the parentheses.
*** p<0.01, ** p<0.05, * p<0.1

Table A7: Property Tax Capitalization: Placebo Tests

VARIABLES	log(Housing Prices)		
	Falsified Cutoff Low Tax Group	Falsified Cutoff High Tax Group	Purchase Restriction
	(1)	(2)	(3)
Median Low Tax Group × Post	-0.0107		
	(0.00895)		
Median High Tax Group × Post		-0.00105	
		(0.0127)	
High Tax Group × Purchase Restriction			0.00746
			(0.00768)
Observations	8,208	1,892	5,043
R-squared	0.958	0.928	0.989
Neighborhood FEs	YES	YES	YES
Year-Month FEs	YES	YES	YES

Note: Column (1) reports the placebo test on sample in the low tax bracket using the median price in the low tax bracket as the cutoff, Column (2) reports the placebo test on sample in the high tax bracket using the median price in the high tax bracket as the cutoff, Column (3) reports the placebo test using the timing of the housing purchase restriction. Neighborhood fixed effects, year-month fixed effects, each neighborhood's distance to the nearest subway stations and district-level population, housing investment and household consumption are controlled for all columns. Standard errors clustered by district are reported in the parentheses.
*** p<0.01, ** p<0.05, * p<0.1